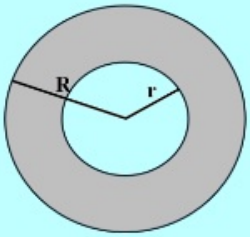
CHA-5.C.3 Volumes of solids of revolution around the x- or y-axis whose cross sections are ring shaped may be found using definite integrals with the washer method.
CHA-5.C.4 Volumes of solids of revolution around any horizontal or vertical line whose cross sections are ring shaped may be found using definite integrals with the washer method.

8-11 & 8-12 Volume with Washer Method

Get Ready...

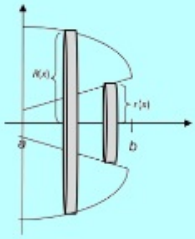
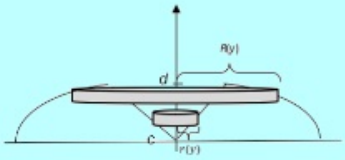
What is an expression for the area of the shape at the right?



$\pi R^2 - \pi r^2$

$\pi[R^2 - r^2]$

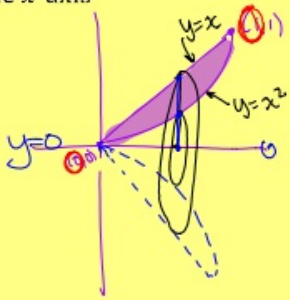
In the last section, we found the volumes of solids using discs. Sometimes when we rotate a given area over a line, we have a "hole" in the solid, so a disc cannot be used to find the area. Instead, we use a washer.

Volume of a Solid of Revolution (Washer Method)	
Horizontal Axis of Revolution	**Vertical Axis of Revolution**
$\text{Volume} = \int_a^b \pi \left[\left[\begin{matrix} \text{Outer} \\ \text{Radius} \end{matrix} \right]^2 - \left[\begin{matrix} \text{Inner} \\ \text{Radius} \end{matrix} \right]^2 \right] dx$ $= \pi \int_a^b [[R(x)]^2 - [r(x)]^2]\, dx$	$\text{Volume} = \int_c^d \pi \left[\left[\begin{matrix} \text{Outer} \\ \text{Radius} \end{matrix} \right]^2 - \left[\begin{matrix} \text{Inner} \\ \text{Radius} \end{matrix} \right]^2 \right] dy$ $= \pi \int_c^d [[R(y)]^2 - [r(y)]^2]\, dy$
	

How to find the Volume of a Solid of Revolution (Washer Method)
1.) Draw the region 2.) Draw the reflection of the region over the axis of revolution (x-axis, y-axis, or another line). 3.) Draw the **washer** that represents the rotation of the area around the axis of revolution. It will be **perpendicular** to the axis of revolution. 4.) Draw the radius of the Washer by drawing **the outer radius and the inner radius** from the edge of the **original area** to the axis of revolution. 5.) Determine the expressions that represent the outer radius and the inner radius. 6.) Use the Volume formula to find the volume.

Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified line. Sketch the region, the solid, and a typical "washer."

Example 1: $y = x,\ y = x^2$ rotated about the x-axis



Outer radius $= x - 0$

inner radius $= x^2 - 0$

$$V = \pi \int_0^1 [x-0]^2 - [x^2-0]^2 \, dx$$

$$= \pi \int_0^1 [x]^2 - [x^2]^2 \, dx$$

$$= \pi \int_0^1 (x^2 - x^4) \, dx$$

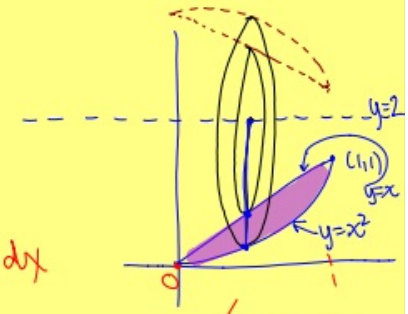
$$= \pi \left[\tfrac{1}{3}x^3 - \tfrac{1}{5}x^5 \right]\Big|_0^1$$

$$= \pi \left(\tfrac{1}{3} - \tfrac{1}{5} - 0 \right) = \boxed{\frac{2\pi}{15}}$$

Example 2: $y = x,\ y = x^2$ rotated about the line $y = 2$

Outer radius $= 2 - x^2$

inner radius $= 2 - x$



$$V = \pi \int_0^1 [2-x^2]^2 - [2-x]^2 \, dx$$

$$= \pi \int_0^1 4 - 4x^2 + x^4 + (4 + 4x - x^2) \, dx \quad \{(2-x^2)(2-x^2)$$

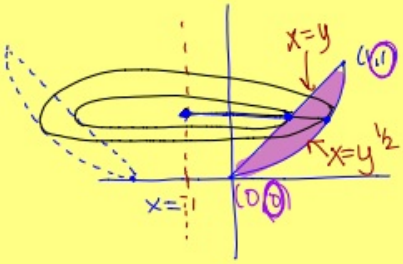
$$= \pi \int_0^1 x^4 - 5x^2 + 4x \, dx$$

$$= \pi \left[\tfrac{1}{5}x^5 - \tfrac{5}{3}x^3 + 2x^2 \right] \Big|_0^1$$

$$= \pi \left(\tfrac{1}{5} - \tfrac{5}{3} + 2 - 0 \right) = \boxed{\frac{8\pi}{15}}$$

Example 3: $y = x,\ y = x^2$ rotated about the line $x = -1$

$$y = x^2$$
$$\sqrt{y} = x$$
$$y^{1/2} = x$$



Outer radius $= y^{1/2} - -1$

Inner radius $= y - -1$

$$V = \pi \int_0^1 \left[y^{1/2} + 1\right]^2 - \left[y + 1\right]^2 \, dy$$

$$= \pi \int_0^1 y + 2y^{1/2} + 1 + \left(-y^2 \mp 2y \mp 1\right) dy$$

$$= \pi \int_0^1 \left(-y^2 - y + 2y^{1/2}\right) dy$$

$$= \pi \left[-\tfrac{1}{3}y^3 - \tfrac{1}{2}y^2 + \tfrac{4}{3}y^{3/2}\right]\Big|_0^1$$

$$= \pi \left[-\tfrac{1}{3} - \tfrac{1}{2} + \tfrac{4}{3} - 0\right] = \boxed{\frac{\pi}{2}}$$